

Epiplexity with lightweight learning algorithms

Experimental design for MLP, CNN and DiT setups

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Epiplexity: information accessible to a bounded learner

Epiplexity is the structural information in data that an observer limited to time T can extract (Finzi et al., 2026).

Minimise a two-part code over models $P \in \mathcal{P}_T$ that run in time T :

$$P^* = \arg \min_{P \in \mathcal{P}_T} \{|P| + \mathbb{E}[-\log P(X)]\}.$$

The minimum splits into structure and randomness:

$$\underbrace{\text{MDL}_T(X)}_{\text{total}} = \underbrace{|P^*|}_{S_T(X): \text{ epiplexity}} + \underbrace{\mathbb{E}[-\log P^*(X)]}_{H_T(X): \text{ time-bounded entropy}}$$

- S_T depends on the bound T and on the class $\mathcal{P}_T$.
- Neural training approximates the minimisation over P .

Random and structural information

Random vs structural information

Low random info, low structural info

```
def is_even(n):  
    if n == 0: return True  
    elif n == 1: return False  
    elif n == 2: return True  
    elif n == 3: return False  
    elif n == 4: return True  
    elif n == 5: return False  
    ...
```



Moderate random info, high structural info

```
def dijkstra(g, s):  
    D = defaultdict(lambda: float('inf'))  
    D[s] = 0; q = [(0, s)]  
    while q:  
        d, u = pop(q)  
        if d >= D[u]:  
            for v, w in g.get(u, []):  
                if (nd := d + w) < D[v]:  
                    D[v] = nd; push(q, (nd, v))  
    return D
```

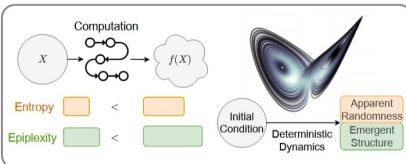


High random info, low structural info

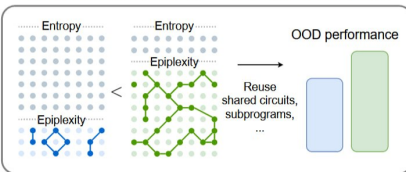
```
APT_KEY = "sk_7af25k1ycP9LmVzz3A"  
USER_ID = "usr_4f8a2c1e9b7d3065"  
BUCKET = "s3://data-8a3f1b-west-prod"  
SAVE_DIR = "/mnt/narc/exp_7f2a/ckpts"  
SAVE_CKPT = True  
DEBUG = False  
SEED = 9284715  
...
```



Information can be created by computation



Structural information → OOD generalization



Source: Fig. 1 of Finzi et al. (2026).

What we test: the estimators, not the architectures

The same estimators run in three setups (DiT: diffusion transformer):

$$X \xrightarrow{\text{setup } a} P_T, \quad a \in \{\text{MLP, CNN, DiT + sampler}\}.$$

Two sources of change have to be told apart:

- *manipulations* inside a setup: data order, pixel permutation, sampler schedule, bound T ;
- the *observer family*: hypothesis class and inductive bias.

Goal: which part of each estimate comes from the data, which from the setup.

Estimators: one learning trajectory, several codes

Model P_i is trained on $z_0, \dots, z_{i-1}$ out of M examples; $\ell_i(z) = -\log P_i(z)$.

Prequential (Finzi et al., 2026): area of the loss curve above the final loss.

$$S_{\text{preq}} \approx \sum_{i < M} [\ell_i(z_i) - \ell_M(z_i)]$$

Requential (Qiu et al., 2026): student s trained on samples of teacher t .

$$S_{\text{req}} \approx \sum_i \text{KL}(P_i^t \parallel P_i^s)$$

EDL (Donoway et al., 2026): floor is the population loss L of the final P_T .

$$\text{EDL} = \mathbb{E} \sum_{i < M} \ell_i(z_i) - M L(P_T)$$

EDL fixes the training algorithm; epiplexity minimises over $\mathcal{P}_T$.

Experimental design: three setups, one manipulation each

Benchmark: elementary cellular automata (ECA), as in Finzi et al. (2026). The rule sets how much learnable structure the data holds.

	MLP	CNN	DiT
Parameters	10^3 – 10^6	10^3 – 10^6	small $\rightarrow$ medium
Prior	weakest	locality	generative, iterative
Optimizer	SGD / Adam(W)	SGD / Adam(W)	Adam(W)
Manipulation	example order	pixel permutation	sampler, #steps

- MLP: no spatial prior; example order is part of the observer.
- CNN: fixed pixel permutation; labels kept, locality removed.
- DiT: the sampler is part of the observer.

Within one comparison the task and T are fixed.

Validation

For each task d , setup a , bound T and seed s we record the training run and compute S_{preq} , S_{req} and EDL.

- **Agreement:** Spearman and Kendall τ between task orderings, within and across setups.
- **Setup dependence:** $\Delta_{ab}(d, T) = S(d, a, T) - S(d, b, T)$. Large Δ with a stable task order: the estimate partly measures the setup.
- **Seed variance**, compared with the spread across setups.
- **Baselines:** final test loss, sample efficiency.

Deliverable: a map of where each estimator is setup-invariant.

Literature

- M. Finzi, S. Qiu, Y. Jiang, P. Izmailov, J. Zico Kolter, A. G. Wilson. *From Entropy to Epiplexity: Rethinking Information for Computationally Bounded Intelligence*. arXiv:2601.03220, 2026.
- S. Qiu, M. Finzi, Y. Zheng, K. Zhang, A. G. Wilson. *Requential Coding: Pushing the Limits of Model Compression with Self-Generated Training Data*. arXiv:2607.11883, 2026.
- E. Donoway, H. Joren, F. Roger, J. Leike. *Excess Description Length of Learning Generalizable Predictors*. arXiv:2601.04728, 2026.